

Lab 1 Analysis

Group: Carl Jonsson & Adam Persson

Activity 1.1 Quality-attribute analysis

FR-1.1 Registration

The first ISO/IEC 25010 quality characteristic we chose was **3.4.4 user error protection**. We chose this one because a user should not be able to register an account with invalid credentials.

The second quality characteristic we chose was **3.4.8 self-descriptiveness**. We chose this one because the system should provide appropriate/necessary information to the user during registration regarding the required format of the input.

FR-2.2 Capacity rule

The first quality characteristic we chose was **3.1.2 functional correctness**. We chose this one because a seeker should not be able to take a job for a provider if the provider already has reached the capacity limit for active bookings.

The second quality characteristic we chose was **3.5.2 availability**. We chose this one to ensure that the provider's current capacity status is available when needed, so seekers are immediately notified when a provider has reached their booking limit and cannot accept any new bookings.

FR-4.3 Pricing

The first quality characteristic we chose was **3.6.2 integrity**. We chose this one because unauthorized users should not be able to change/manipulate the price of a job/booking.

The second quality characteristic we chose was **3.1.2 functional correctness**. We chose this one because the price must be calculated accurately based on trust tiers and the job's duration.

Testable quality requirement

A seeker with a trust tier of **NEW** applies for a **DOG_WALK** booking with a duration of 9 hours. The price should be calculated as follows:

- **NEW** tier: Platform fee of 15%
- **DOG_WALK** base rate: 80 SEK/hour
- **overnight surcharge**: 20%

Expected price: $80 \times 9 = 720 \rightarrow 720 + 20\% = 864 \rightarrow 864 + 15\% = 993,6$ SEK

The expected outcome is that the system should calculate a price of 993.6 SEK for the user.

Activity 1.2 Bug analysis

The human error is a logic mistake, and the fault is usage of the wrong operator. The if-statement for rule number 2 uses ">" instead of ">=" as shown below.

```
if (seekerActive > seeker.getMaxConcurrentBookings()) { ... }
```

The test level that should have caught this is the "Component (unit)" level during white box testing.

Activity 2.1 Equivalence Partitioning

Email format/length

Input Field	Partition	Representative Value	Expected Outcome
Email	Valid, proper format	example@example.com	Accepted
Email	Invalid, missing @	example.com	Rejected
Email	Invalid, multiple @	test@@example.com	Rejected
Email	Invalid, missing .	test@examplecom	Rejected
Email	Invalid, missing local part	@example.com	Rejected
Email	Invalid, too long local part	a.....b@example.com	Rejected

Display name format/length

Input Field	Partition	Representative Value	Expected Outcome
Display name	Valid, proper length and valid characters	Adam-Persson	Accepted
Display name	Invalid, too short (<2 chars)	A	Rejected
Display name	Invalid, too long (>40 chars)	Aaaaaaaaaaaaaaaaaaaa...	Rejected
Display name	Invalid, contains digits	Adam123	Rejected
Display name	Invalid, contains special characters	Adam:)	Rejected

Phone number format/length

Input Field	Partition	Representative Value	Expected Outcome
Phone number	Valid (Swedish format, 10 digits, starts with 07)	0731231234	Accepted
Phone number	Valid (International, 12 digits, starts with +46)	+46731231234	Accepted
Phone number	Invalid, swedish format, too long	07312312312	Rejected

Input Field	Partition	Representative Value	Expected Outcome
Phone number	Invalid, swedish format, too short	073123123	Rejected
Phone number	Invalid, international, too long	+467312312345	Rejected
Phone number	Invalid, international, too short	+4673123123	Rejected

Wallet top-up amount

Input Field	Partition	Representative Value	Expected Outcome
Wallet top-up	Valid, top-up between 10.00 and 5 000.00 SEK inclusive	1000.00	Accepted
Wallet top-up	Invalid, top-up below 10.00 SEK	9.99	Rejected
Wallet top-up	Invalid, top-up above 5 000.00 SEK	5 001.00	Rejected
Wallet top-up	Valid, maximum resulting balance $\leq$ 20 000.00 SEK	20 000.00	Accepted
Wallet top-up	Invalid, maximum resulting balance $>$ 20 000.00 SEK	20 001.00	Rejected
Wallet balance	Invalid, balance turns negative	- 10.00	Rejected

Activity 2.2 Boundary Value Analysis

Purpose	Value	Expected Result
Just below lower boundary	9.99	Rejected
Lower boundary	10.00	Accepted
Just above lower boundary	10.01	Accepted
Just below upper boundary	4 999.99	Accepted
Upper boundary	5 000.00	Accepted
Just above upper boundary	5 000.01	Rejected

Activity 2.3 Decision table (trust tier → limits)

Condition / Expected result	NEW	VERIFIED	TRUSTED	PRO_SITTER
Max concurrent bookings	1	3	5	10
Platform fee	15%	12%	8%	5%